

VAVA'U, Tonga

WMO: 917790

Lat: 18.583S

Lon: 173.967W

Elev: 230

StdP: 14.57

Time Zone: 13.00 (E13)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	60.8	63.6	55.7	66.7	69.1	57.6	71.4	69.4	20.5	76.5	18.5	75.2	0.1	140	0.323

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	8.8	86.4	79.3	86.1	79.1	85.4	78.6	80.7	84.7	80.0	84.0	79.4	83.4	9.9	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.2	152.8	83.5	78.9	151.5	83.3	78.5	149.3	82.9	44.6	85.8	44.0	84.4	43.4	83.3	83.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.3	17.0	15.6	57.1	87.8	2.1	1.1	55.6	88.6	54.4	89.2	53.3	89.8	51.8	90.6		
(5)			55.6	82.1	1.4	0.7	54.6	82.6	53.8	83.0	53.0	83.4	52.0	83.8	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	76.9	79.6	80.3	79.8	78.4	76.5	75.0	73.7	73.7	74.6	75.8	77.0	78.7
	DBStd	3.42	1.96	1.97	1.89	2.29	2.65	2.62	3.00	2.84	2.93	2.80	2.75	2.38
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	9819	919	848	923	853	820	751	734	734	739	800	810	888
	CDD65	4344	454	428	458	403	355	301	269	269	289	335	360	423
	CDH74	32610	4373	4255	4429	3510	2385	1501	1103	1096	1454	1986	2745	3773
CDH80	5406	991	1062	1034	630	209	87	39	49	96	164	355	690	
Wind	WSAvg	7.6	7.7	8.0	6.3	7.2	7.6	7.6	7.4	7.6	7.8	8.1	7.9	8.0
Precipitation	PrecAvg	85.4	11.3	8.8	10.6	8.7	5.9	5.4	4.0	3.7	5.6	6.7	5.7	8.0
	PrecMax	134.8	24.3	19.2	24.4	23.0	14.6	20.8	9.2	13.2	16.7	16.8	18.8	20.9
	PrecMin	42.0	1.8	2.5	2.5	1.9	1.1	0.0	0.6	0.4	0.0	0.0	0.2	1.3
	PrecStd	20.2	7.3	3.9	5.2	5.4	3.7	5.3	2.4	2.8	4.4	4.4	4.2	5.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	87.6	88.2	87.3	86.3	84.5	83.9	82.5	82.6	83.8	84.1	85.1	86.3
		MCWB	80.1	80.3	80.0	79.0	77.8	77.8	76.5	75.8	77.3	76.8	77.7	78.8
	2%	DB	86.2	86.9	86.2	85.4	83.0	82.2	80.9	81.1	82.3	82.7	84.3	85.4
		MCWB	79.0	79.8	79.3	78.3	76.8	76.6	75.3	74.8	75.4	75.7	77.4	78.2
	5%	DB	85.6	86.0	85.6	84.3	82.3	80.8	80.0	80.4	80.9	82.0	83.1	84.4
		MCWB	78.7	79.3	78.9	77.5	76.3	75.3	74.5	74.3	74.3	75.4	76.5	77.5
	10%	DB	84.3	84.7	84.5	83.4	80.9	79.4	78.8	78.9	79.7	80.7	82.3	83.8
		MCWB	78.0	78.6	78.3	76.9	75.2	74.3	73.3	73.2	73.5	74.7	75.9	77.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.0	81.6	81.5	80.2	79.2	78.9	78.1	77.7	78.4	78.3	79.7	80.7
		MCDB	84.9	85.6	85.1	84.1	82.4	82.0	80.8	80.6	82.3	81.6	83.1	84.0
	2%	WB	80.2	80.7	80.5	79.2	78.3	77.8	77.0	76.5	76.7	77.3	78.7	79.2
		MCDB	84.1	85.0	84.4	83.3	81.5	80.8	79.9	79.7	80.3	81.0	82.4	83.1
	5%	WB	79.4	80.0	79.8	78.7	77.6	76.8	75.9	75.6	75.8	76.5	77.9	78.7
		MCDB	83.3	84.2	83.7	82.9	80.8	79.7	78.8	78.8	79.3	80.2	81.7	82.8
	10%	WB	78.9	79.3	79.1	78.1	76.9	75.8	74.9	74.6	74.8	75.7	77.1	78.0
		MCDB	82.8	83.3	83.1	82.1	80.1	78.7	77.7	77.8	78.3	79.4	80.9	82.1
Mean Daily Temperature Range	5% DB	MDBR	9.2	8.8	9.5	9.7	9.1	8.5	9.2	9.4	9.5	9.0	9.5	9.3
		MCDBR	10.9	10.3	10.2	11.2	10.0	10.1	10.5	10.8	10.6	10.1	10.4	10.0
		MCWBR	5.8	5.3	5.4	6.1	5.8	6.3	6.6	6.5	6.2	5.7	5.9	5.2
	5% WB	MCDBR	9.1	8.8	9.2	9.7	8.8	8.8	8.5	8.9	8.8	8.9	9.1	9.2
MCWBR		5.4	4.9	5.4	5.6	5.8	5.9	6.1	6.1	6.0	5.6	5.9	5.3	
Clear-Sky Solar Irradiance	taub	0.384	0.379	0.373	0.394	0.382	0.384	0.366	0.375	0.397	0.382	0.380	0.386	
	taud	2.626	2.644	2.679	2.594	2.610	2.580	2.625	2.599	2.537	2.582	2.610	2.600	
	Ebn at Noon	305	303	298	279	271	264	273	281	285	298	304	304	
	Edh at Noon	32	31	29	30	28	27	27	29	33	33	33	33	
All-Sky Solar Radiation	RadAvg	1994	1896	1815	1522	1332	1207	1312	1488	1686	1879	1972	2031	
	RadStd	208	199	125	156	86	104	72	96	114	134	152	145	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page